Representing Vectors as Diagrams

Exam code: 0607

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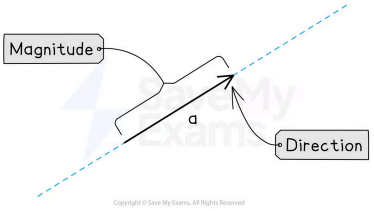
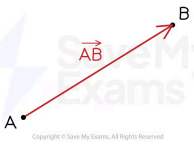
Vector diagrams

How can I represent a vector visually?

- A **vector** has **both a size (magnitude) and a direction**
  - You need to draw a **line** to show the **size of the vector**
  - You also need to draw an **arrow** to show the **direction of the vector**



- Vectors are written in **bold** when typed to **show** that they are a vector and not a scalar
  - When writing a vector in an exam you should **underline** the letter to show it is a vector
  - $\boldsymbol{a}$ when **typed** and $\underline{a}$ when **handwritten**
    - You will **not lose marks** if you forget to underline vectors
- If a vector **starts at *A*** and **ends at *B*** we can write it as $\overrightarrow{AB}$
  - Here the arrow will **point toward *B***
  - Vector $\overrightarrow{BA}$ will have the **same length** but **point toward *A***



How do I draw a vector on a grid?

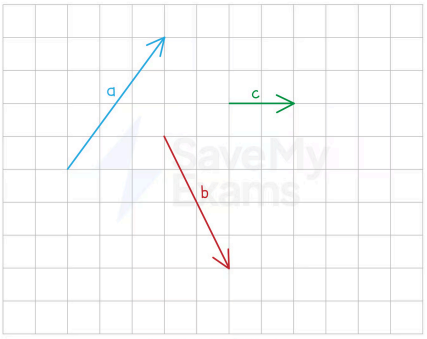
- You can draw a vector **anywhere** on a grid

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- To draw the vector $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$
  - **Pick** a point on the grid and draw a **dot** there
  - Count 3 units to the right and 4 units up and **draw another dot**
  - Draw a **line between the two dots**
  - Put an **arrow** on the line **pointing toward the second dot**

- Look out for **negatives** and **zeroes**
  - $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ goes **2 to the right** and **4 down**
  - $\mathbf{c} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ goes **2 to the right** but **does not go up or down**



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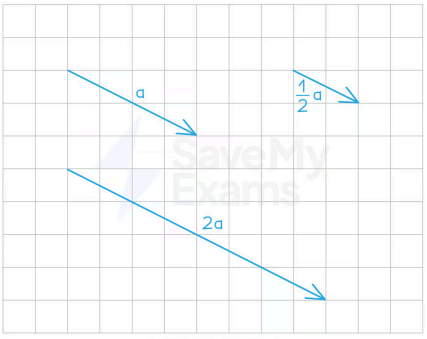
What happens when I multiply a vector by a scalar?

- When you **multiply** a vector by a **positive scalar**:
  - The **direction** stays the **same**
  - The **length** of the vector is **multiplied by the scalar**

- For example, $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$
  - $2\mathbf{a} = \begin{pmatrix} 8 \\ -4 \end{pmatrix}$ will have the **same direction** but **double the length**
  - $\frac{1}{2}\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ will have the **same direction** but **half the length**



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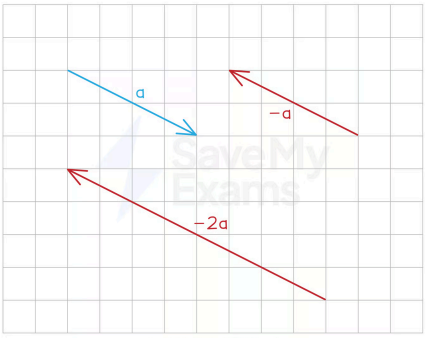
- When you **multiply** a vector by a **negative scalar**:

- For example, $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$
  - $-\mathbf{a} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$ will be in the **opposite direction** and its **length will be the same**
  - $-2\mathbf{a} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$ will be in the **opposite direction** and its **length will be doubled**



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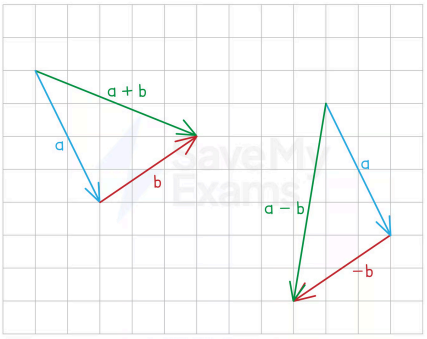
What happens when I add or subtract vectors?

- To **draw** the vector $\mathbf{a} + \mathbf{b}$
  - Draw the vector $\mathbf{a}$
  - Draw the vector $\mathbf{b}$ starting at the endpoint of $\mathbf{a}$
  - Draw a line that **starts at the start** of $\mathbf{a}$ and **ends at the end** of $\mathbf{b}$
- To **draw** the vector $\mathbf{a} - \mathbf{b}$
  - Draw the vector $\mathbf{a}$
  - Draw the vector $-\mathbf{b}$ starting at the endpoint of $\mathbf{a}$
  - Draw a line that **starts at the start** of $\mathbf{a}$ and **ends at the end** of $-\mathbf{b}$



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Worked Example

The points *A*, *B* and *C* have coordinates (-4, 2), (2, 4) and (3, -4), respectively.

Answer:

Start by drawing the three vectors onto a grid

From *A* to *B*, it is 6 to the right and 2 up

$$\overrightarrow{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

From *A* to *C*, it is 7 to the right and 6 down

$$\overrightarrow{AC} = \begin{pmatrix} 7 \\ -6 \end{pmatrix}$$

From C to *B*, it is 1 to the left and 8 up

$$\overrightarrow{CB} = \begin{pmatrix} -1 \\ 8 \end{pmatrix}$$

(b)

Without using any calculations, explain why $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Answer:

The vector goes from A to B, then from B to C, then from C back to A

The vector returns to its starting point

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